Yeast 001 — the script

The spoken words of the supercut, in order — 6:51. Transcribed from the cut itself.

Cold open — 0:00

> This is yeast bubbling in action. And this is me, way too excited about yeast.
>
> You don't actually need sugar to make your yeast grow. You just need time.
>
> Hi, I'm Katie. I'm a food scientist, and this is my deep dive into yeast.

1. Yeast is old — 0:13

> Yeast is really old. We've had it around for at least a few thousand years. We've found it in archaeological records in Egypt. We have found it in archaeological records in parts of Asia and in Europe.
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> It's being passed between brewers and bakers. It's being collected from the air. It's being collected from outside of grapes and introduced into wine. It's moving around. It's passed from person to person, baker to brewer, brewer to baker. It's in mead. It's in wine. It's in beer. It's in our sourdough starters. It's just all around.
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> They may not know what they're breeding, but they're being selective about it. If it's a dud batch, they're not passing it on to their friend.
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> And so what we're getting is selective breeding of good yeast — yeast that ferments, that makes the bread bubble, makes the alcohol happen — is being passed around. And that is what we've used for years and years and years.

2. Louis Pasteur — 1:02

> It wasn't until we meet Louis Pasteur that we have an inkling of what's making one successful batch work, and what's being passed on between these bakers and brewers — has a gooseneck experiment, and disproves spontaneous generation, and says there are microbes, little bugs that we can't see, all around the world.
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> If you don't know about the Louis Pasteur experiment: what he does is he takes a glass bulb and stretches out a neck on it, that is open over here, but it has a gooseneck shape. And he heats it up to kill off any bacteria that's on the inside, with some food that he has cooked, that he's put inside of this gooseneck bottle. And so if there's any truth to spontaneous generation, the food will just break down. There will be no problem with it doing what food has always done. But if the food breaks down because of germ theory — what he was thinking: germs can't get in through the gooseneck bottle. It falls down. It gets stuck in the downspout, right here. So nothing will happen to the food. It won't break down. It won't do anything. And that's what happened. And so Louis Pasteur proves that there is no spontaneous generation — that there is something in the air: bacteria, yeast, living organisms that are getting into our food and making changes.

3. One bug — 2:13

> We meet our *Saccharomyces cerevisiae*. And this is the most common form of yeast that we use today. There are other yeasts that are used in sourdoughs, but for the most part, when you get to industrial bakers and you get to home bakers and you get to brewers, they're all using the same little bug: *Saccharomyces cerevisiae*.
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> It is like 90% of all the yeast that we use is this one bug. All over — we use it everywhere, from our home bakeries to the brewers down the road to the vineyards in California and France. Everywhere. *Saccharomyces cerevisiae*. Same guy, again and again and again.

4. What's on the shelf — 2:51

> I have marketing for yeast for mead. It's *Saccharomyces cerevisiae*. I have instant dry yeast. It's *Saccharomyces cerevisiae*. I have active dry yeast with a little sugar in it. Guess what? *Saccharomyces cerevisiae*. Fresh yeast. *Saccharomyces cerevisiae*. I even have what could be the start of a sourdough starter. Here, it's a pre-ferment. It is yeast, *Saccharomyces cerevisiae*, mixed with flour and water.
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> And so someone with a marketing degree, with a business degree, is saying: hey, let me sell to you. I know what you need. It's this. But it's really all just the same *Saccharomyces cerevisiae*.

5. Access and need — 3:23

> Why do we have so many different names for it? Why do we have so many different formats for it? The answer is access and need.
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> You have the at-home baker in the middle of a farm in Kansas that needs something every once in a while, and doesn't want to have to drive two hours back and forth to a grocery store every couple days to get fresh yeast. They need something on hand that they can keep, that won't die off on them as frequently. So you have fresh yeast and dry yeast, and you have active dry and instant.
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> But in the processing of the active dry yeast and the instant yeast, the yeast gets a little damaged. It helps dry out the bread a little bit faster. And to be honest, an at-home baker might not care. You might not notice. They consume the bread within a couple days, because they're only making one loaf at a time. Because it might not be worth driving a couple hours to a grocery store to get something to have a bread that's a little bit stretchier that lasts a day longer.

6. Budding — 4:15

> So you have a mother cell that grows, and buds on itself a daughter cell. And when that daughter cell is big and identical to the mother cell, it uses mitosis and splits into two cells. The yeast will continue to grow and reproduce itself when it's in an environment that has lots of nutrients for it. And those nutrients are sugars and starches. It breaks down starch into sugar.

7. What it eats — 4:36

> The yeast will use simple sugars and break it down with the enzyme zymase into CO_2 and alcohols. And the CO_2 and the alcohol are the byproducts that we're looking for as humans — either in alcohol production, or in helping our bread rise with the CO_2, giving the gas that we will eventually bake off.
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> And you also have your CO_2. That's where you get your bubbly wines, like your champagne.

8. Temperature — 5:00

> Yeast is looking for a right environment for it to thrive and to grow, and that environment has lots of sugars and starches, and is in a temperature between 50 degrees, where it kind of shuts down and goes dormant, and 140 degrees, when it's completely killed off. There's a nice sweet spot right in the middle — about 93 degrees Fahrenheit and 100 plus a couple degrees Fahrenheit. And in that sweet temperature spot, it will grow and thrive.

9. Faster and slower — 5:29

> So when we add salt and fat and other spices to our bread doughs, what we're doing is we're putting items in the way of the yeast and its food. And so it's slowing down the process in which the yeast can digest the sugars that it wants for growing, and that will then slow our proofing time. The availability of sugar will speed up the proofing time.
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> This is why in active dry yeast they recommend that you add a little bit of sugar, to kind of start the yeast up and going and getting it bubbling. You don't have to actually do that. You can just let it sit a little longer. Yeast itself just really wants flour and water. You can do that sometimes by getting a pre-ferment. This is what I have going right here, and it's already starting to bubble up, and it's just flour and water and yeast, and the yeast is in a happy spot.
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> Adding sugar does speed it up, but then you end up with a sweeter dough. And if that is not the final product that you want, you do not actually need to add sugar to your dough to get the yeast to grow.

10. The same little bug — 6:26

> Use what's available. It's all the same. It can be played with. Yeast is not something that has to be a hard and fast rule. It feels like a mystery, but really all it is is the same little *Saccharomyces cerevisiae* bug, again and again and again and again.
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> All the little bug wants is to go and eat sugars and fart — produce carbon dioxide and ethanol. It just wants to live a happy life, and it's not all that complicated.